ML Assignment 2

Should an AI Agent
Call a Tool, or Refuse?

can_answer: 0 | 1

3,491
Agent Tasks

33
Features

2
Classes

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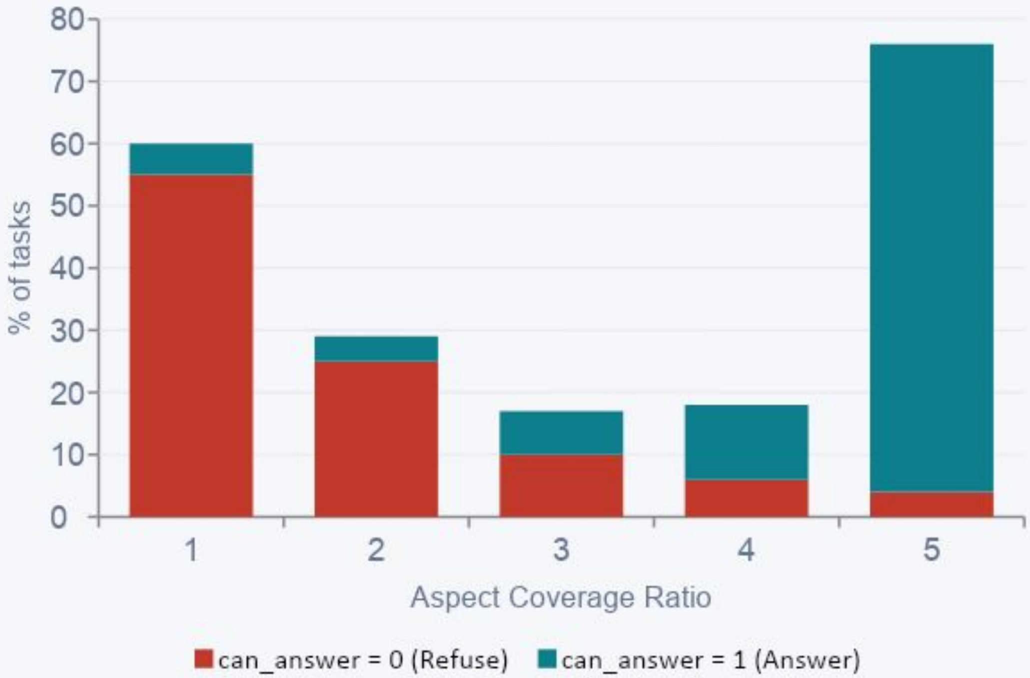
EDA Insight

Aspect Coverage Ratio Separates the Classes



Tasks that can be answered overwhelmingly have ratio near 1.0 — tools match the query topics.

Refusal tasks cluster at 0 — available tools are irrelevant to the query.

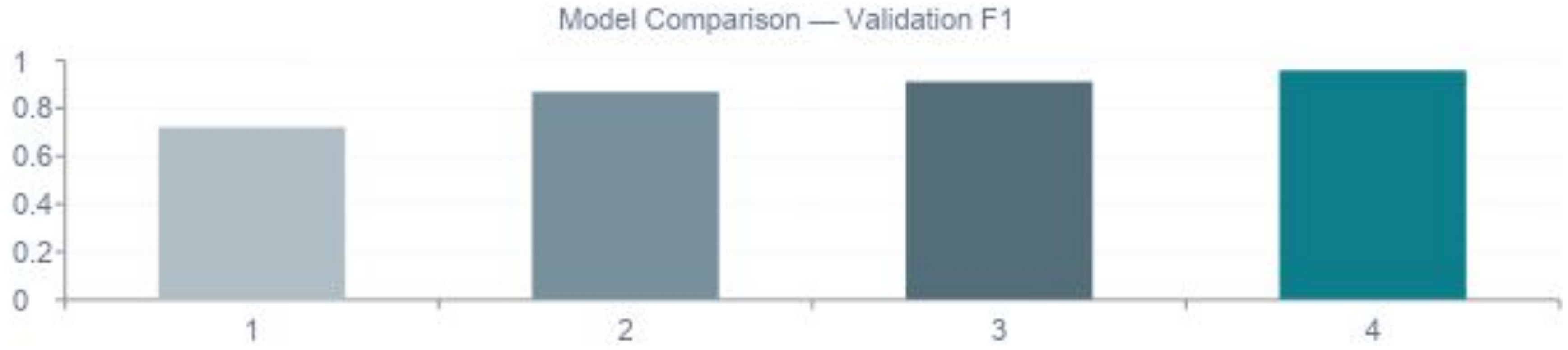
This single feature achieves strong class separation and becomes the most important model feature.

Best Classification Model

Random Forest — Test Set Results

95%	96%	98%	97%
Accuracy	Precision	Recall	F1-Score

`Hyperparameters:  n_estimators = 200  ·  max_depth = 20`



False Positives Are the Costliest Mistake

False Positive

`Predict CAN answer · Actually CANNOT`

- Agent calls a tool that cannot help
- May perform harmful actions (e.g. send, book, delete)
- Wastes resources and erodes user trust
- Harder to recover from in production

Most Dangerous

False Negative

`Predict CANNOT answer · Actually CAN`

- Agent refuses a solvable request
- User experience suffers — unhelpful AI
- No harmful side-effect
- User can rephrase and retry

Recoverable

Preference: A conservative model (high precision) is preferred in production to prevent unsafe tool calls.

Clustering Insight

Tasks Cluster Naturally by Tool Relevance

K-Means (k=2) | Agglomerative (Ward)

Cluster 0

Low-Match / Refusal

Answer Rate	**~5%**
Aspect Coverage	**~0.05**
Top Domain	**mixed**
Error Profile	**Mostly TN**
Complexity	**low–medium**

Cluster 1

High-Match / Answerable

Answer Rate	**~98%**
Aspect Coverage	**~0.92**
Top Domain	**travel / other**
Error Profile	**Mostly TP**
Complexity	**low**

Final Recommendation & Limitation

Dataset Limitation · Suggested Improvement

Dataset Limitation

The can_answer label is synthetically derived from benchmark construction rules — not from new human annotations. Models may learn patterns specific to the benchmark artifacts rather than truly generalising to real-world tool-use reasoning.

Suggested Improvement

Replace strict keyword-based aspect matching with embedding-based cosine similarity between the query and tool descriptions. This would capture semantic equivalences (synonyms, paraphrases) that keyword rules miss, creating far richer and more robust features.

Best Model: Random Forest · Main Metric: F1 · Test F1: ~0.97 · Main Error: False Positives